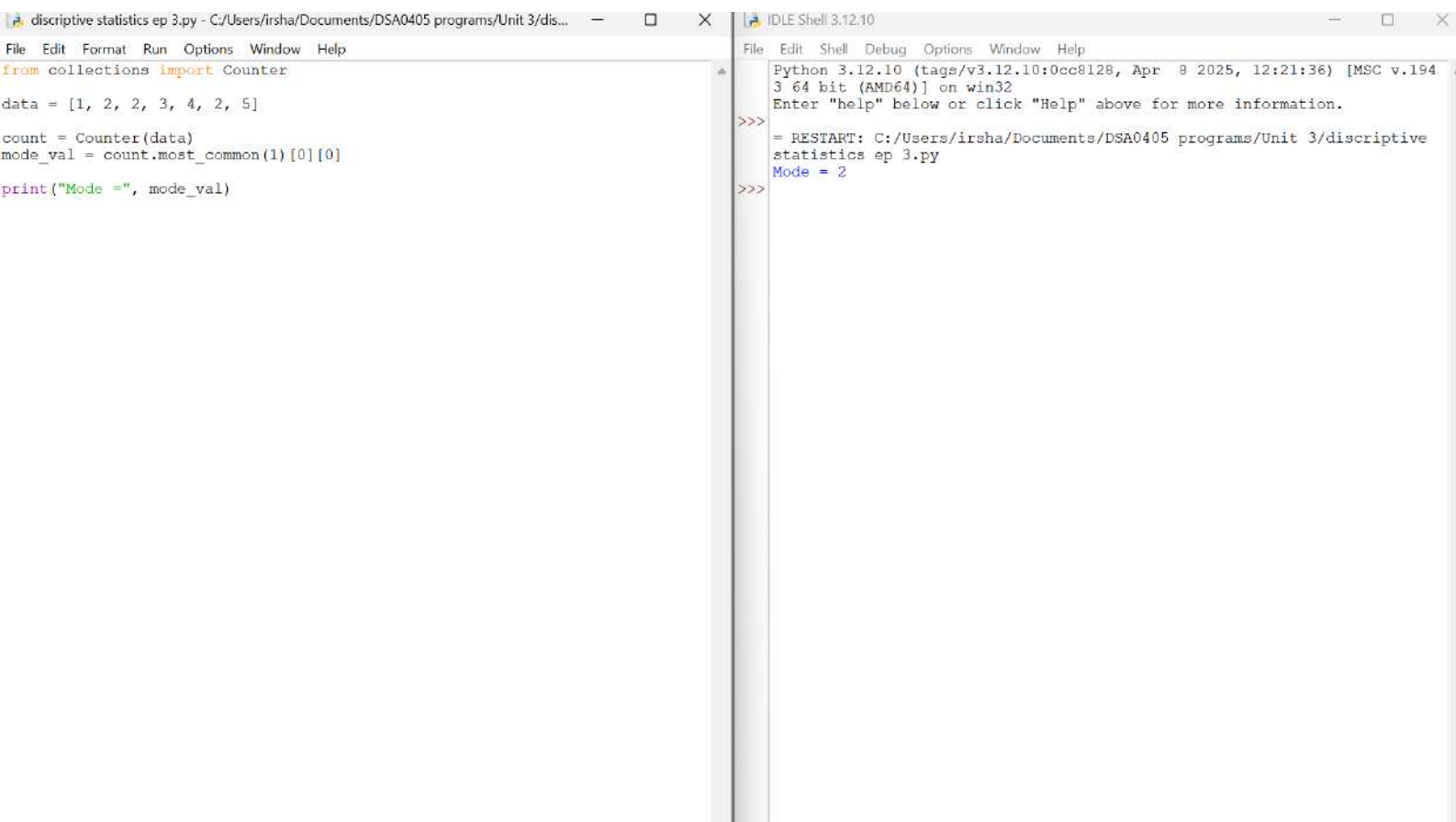


```
discriptive statistics ep 1.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/dis...
File Edit Format Run Options Window Help
data = [10, 20, 30, 40, 50]
mean_val = sum(data) / len(data)
print("Mean =", mean_val)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/discriptive
statistics ep 1.py
Mean = 30.0
>>>
```

```
discriptive statistics ep 2.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/dis...
File Edit Format Run Options Window Help
data = [40, 10, 30, 50, 20]
data.sort()
n = len(data)
if n % 2 == 0:
    median_val = (data[n//2 - 1] + data[n//2]) / 2
else:
    median_val = data[n//2]
print("Median =", median_val)

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/discriptive
statistics ep 2.py
Median = 30
>>>
```



The image shows a screenshot of a Python IDLE Shell window. The window title is "discriptive statistics ep 3.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/dis...". The menu bar includes File, Edit, Format, Run, Options, Window, and Help. The code in the editor is as follows:

```
from collections import Counter

data = [1, 2, 2, 3, 4, 2, 5]

count = Counter(data)
mode_val = count.most_common(1)[0][0]

print("Mode =", mode_val)
```

The IDLE Shell window title is "IDLE Shell 3.12.10". The menu bar includes File, Edit, Shell, Debug, Options, Window, and Help. The output in the shell is:

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/discriptive
statistics ep 3.py
Mode = 2
>>>
```

range.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/range.py (3.12.10)

File Edit Format Run Options Window Help

```
data = [15, 8, 22, 42, 10]
```

```
range_val = max(data) - min(data)
```

```
print("Range =", range_val)
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>> ===== RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/range.p  
y -----  
Range = 34  
>>>
```

```
standard deviation.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/standard...
File Edit Format Run Options Window Help
import math
data = [2, 4, 4, 4, 5, 5, 7, 9]
mean_val = sum(data) / len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
std_dev = math.sqrt(variance_val)
print("Standard Deviation =", std_dev)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/standard dev
iation.py
Standard Deviation = 2.0
>>>
```

```
using built-in.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using built-in...
File Edit Format Run Options Window Help
import statistics

data = [12, 15, 15, 18, 22, 25]

print("Mean =", statistics.mean(data))
print("Median =", statistics.median(data))
print("Mode =", statistics.mode(data))
print("Variance =", statistics.variance(data))
print("Standard Deviation =", statistics.stdev(data))

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using built
-in.py =
Mean = 17.833333333333332
Median = 16.5
Mode = 15
Variance = 23.766666666666666
Standard Deviation = 4.875106836436168
>>>
```

```
using numpy library.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using n...
File Edit Format Run Options Window Help
import numpy as np

data = np.array([10, 20, 20, 30, 40])

print("Mean =", np.mean(data))
print("Median =", np.median(data))
print("Variance =", np.var(data))
print("Standard Deviation =", np.std(data))
print("Range =", np.ptp(data))

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8120, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using numpy
library.py
Mean = 24.0
Median = 20.0
Variance = 104.0
Standard Deviation = 10.198039027185569
Range = 30
>>>
```

```
using pandas.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using pandas....
File Edit Format Run Options Window Help
import pandas as pd

df = pd.DataFrame({
    "Values": [10, 20, 30, 40, 50, 50]
})

print("Mean =", df["Values"].mean())
print("Median =", df["Values"].median())
print("Mode =", df["Values"].mode().tolist())
print("Variance =", df["Values"].var())
print("Standard Deviation =", df["Values"].std())
print("Range =", df["Values"].max() - df["Values"].min())
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using pand
as.py ==
Mean = 33.333333333333336
Median = 35.0
Mode = [50]
Variance = 266.6666666666667
Standard Deviation = 16.32993161855452
Range = 40
>>>
```


using scipy (mode).py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using sci...
File Edit Format Run Options Window Help

from scipy import stats

data = [5, 2, 5, 3, 5, 4]

mode_result = stats.mode(data, keepdims=True)

print("Mode =", mode_result.mode.item())
print("Count =", mode_result.count.item())

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/using scipy
(mode).py
Mode = 5
Count = 3

>>>

variance.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/variance.py (3.12.10)

File Edit Format Run Options Window Help

data = [2, 4, 4, 4, 5, 5, 7, 9]

mean_val = sum(data) / len(data)

variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)

print("Population Variance =", variance_val)

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

==== RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/variance
.PY ====

Population Variance = 4.0

>>>